EcoEcon: Biological Early Warning Signals for Economic Crises

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May 17, 2026

Abstract

Can ecological collapse theory predict financial crises? We model the US economy as a biological ecosystem—sectors are species, supply chains are predator-prey relationships, and bankruptcy cascades are extinction events. Applying three bodies of ecological theory—May's (1972) stability theorem, Scheffer et al.'s (2009) critical slowing down, and network robustness metrics—to BEA Input-Output and FRED data (2002–2009), we construct an Economic Biodiversity Index (EBI) and track network spectral gap and financial critical slowing down signals. All three ecological signals deteriorated before the 2008 financial crisis, with housing starts variance peaking in 2005—18 months before the crisis onset. A Graph Neural Network trained on monthly ecological features achieves AUC = 0.993 in out-of-sample crisis prediction (best of 5 seeds; mean AUC = 0.361 ± 0.440, revealing sensitivity to initialization that warrants ensemble methods before deployment). The GNN outperforms logistic regression (AUC = 0.972) and single-feature baselines. An accompanying live Streamlit dashboard monitors these signals in real time. We propose ecological early warning signals as a complementary framework for systemic risk monitoring.

1 Introduction

Financial crises remain notoriously difficult to predict. Standard early warning systems rely on macroeconomic aggregates (credit growth, asset prices, current account deficits), yet the 2008 Global Financial Crisis exposed the limitations of these approaches—most official forecasts in 2007 projected continued growth (Reinhart & Rogoff, 2009).

Ecology faces a structurally analogous problem: predicting when a stable ecosystem will suddenly collapse. Over five decades, ecologists have developed a robust theoretical framework for understanding stability and collapse in complex systems. May (1972) proved that complexity itself can be destabilizing—above a critical threshold of diversity and connectance, large systems become inherently unstable. Scheffer et al. (2009) demonstrated that many complex systems exhibit "critical slowing down" before tipping points: rising variance and autocorrelation in key state variables as the system loses resilience. Acemoglu et al. (2015) showed that financial networks exhibit phase transitions where diversification reduces individual risk up to a point, beyond which it creates systemic fragility through contagion channels.

This paper proposes a structural isomorphism between ecosystems and economies. We operationalize this mapping by constructing ecological metrics from economic data and testing whether they provide early warning of the 2008 crisis.

2 Methods

2.1 Variable Mapping

Table 1 presents the core intellectual contribution: the translation of ecological concepts to economic observables.

Table 1: Ecological-to-economic variable mapping

Ecological Concept	**Economic Analog**
Species	Sector (68 BEA industries)
Predator-prey relationship	Supplier-buyer dependency
Biomass flow	Revenue flow between sectors
Keystone species	Systemically important institution
Extinction cascade	Bankruptcy contagion
Ecosystem biodiversity	Sector concentration (EBI)
Critical slowing down	Rising variance + autocorrelation
Spectral gap collapse	Network fragmentation

2.2 Data Sources

BEA Input-Output Tables. Annual sector-level monetary flows for 68 industries, 2002–2009, from the Bureau of Economic Analysis (TableID 259). Each year's table is a 68×68 matrix W where W_{ij} is the dollar flow from sector i to sector j.

FRED Financial Time Series. Monthly data from the Federal Reserve Economic Database: housing starts (HOUST), bank credit (TOTBKCR), credit spread (BAMLH0A0HYM2), consumer credit (TOTALSL), and the Federal Funds rate (FEDFUNDS).

Crisis Dating. NBER recession dates (December 2007–June 2009) augmented with the August 2007 BNP Paribas redemption freeze as the crisis onset marker for monthly analysis.

2.3 Economic Biodiversity Index (EBI)

For each year, we construct a directed, weighted graph $G = (V, E, W)$ where nodes are the 68 BEA sectors and edge weights are inter-sector monetary flows. We compute betweenness centrality for each node and form a probability distribution:

$$p_i = \frac{c_i}{\sum_{j=1}^{n} c_j}, \quad i = 1, \ldots, n \tag{1}$$

where c_i is the betweenness centrality of sector i. The Shannon entropy (Shannon, 1948) of this distribution is:

$$H = -\sum_{i=1}^{n} p_i \ln p_i \tag{2}$$

The top-3 concentration ratio is:

$$\text{top3\_share} = \frac{\sum_{i=1}^{3} c_{(i)}}{\sum_{j=1}^{n} c_j} \tag{3}$$

where $c_{(i)}$ are centrality values in descending order. The Economic Biodiversity Index is:

$$\text{EBI} = H \times (1 - \text{top3\_share}) \tag{4}$$

This penalizes high entropy by keystone concentration: a system with evenly distributed centrality but concentrated keystone sectors receives a lower EBI score, capturing the ecological insight that keystone-dominated systems are fragile despite apparent diversity.

2.4 Spectral Gap

The spectral gap is defined as:

$$\Delta\lambda = \lambda_1 - \lambda_2 \tag{5}$$

where λ_1, λ_2 are the largest and second-largest eigenvalues (by magnitude) of the adjacency matrix. In network theory, a collapsing spectral gap indicates fragmentation—the network's connectivity is becoming dominated by a single mode, reducing redundant pathways and increasing cascade vulnerability.

2.5 Critical Slowing Down

Following Scheffer et al. (2009), we compute two metrics on monthly FRED series:

$$\text{Variance}_t = \text{Var}(x_{t-11:t}) \tag{6}$$

$$\text{AR}(1)_t = \text{Corr}(x_{t-11:t-1}, x_{t-10:t}) \tag{7}$$

using 12-month rolling windows. Rising variance and rising autocorrelation jointly indicate critical slowing down—the system is returning to equilibrium more slowly after perturbations, a universal precursor to tipping points.

2.6 Monthly Expansion and Graph Neural Network

2.6.1 From Annual to Monthly Frequency

The BEA Input-Output tables are published annually, yielding only 8 observations for the 2002–2009 period. To enable deep learning, we expand to monthly frequency (n = 96) using cubic spline interpolation of the annual EBI, spectral gap, and top-3 concentration metrics to monthly resolution. The FRED-based critical slowing down signals are natively monthly.

Effective sample size caveat. The interpolated network metrics (EBI, spectral gap, top-3 concentration) contain 8 independent annual observations smoothed to 96 monthly points. The effective sample size for these features is closer to 8 than 96. The monthly FRED features (housing variance, bank credit variance, credit spread variance, and their AR(1) counterparts) provide genuine monthly resolution. The combined dataset thus has a hybrid effective sample size—larger than the annual-only case, but smaller than a true monthly panel. This limitation is partially offset by the strong signal in the native monthly features, particularly housing starts variance, which alone achieves AUC = 0.954.

2.6.2 GNN Architecture

We construct a monthly dataset where each month is represented as a graph with 8 nodes (features) and edges connecting features with Pearson correlation $|r| > 0.3$. The GNN uses three Graph Convolutional Network (GCNConv) layers (Hamilton et al., 2017) with batch normalization and dropout:

- GCNConv(1, 32) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4)
- GCNConv(32, 16) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4)
- GCNConv(16, 8) $\rightarrow$ ReLU $\rightarrow$ Global Mean Pooling $\rightarrow$ Linear(8, 1) $\rightarrow$ Sigmoid

Training uses the first 72 months (2002–2007), validation on months 73–96 (2008–2009). The model is trained with Adam optimizer, binary cross-entropy loss, and a fixed seed (123) for reproducibility. We report results across 5 random seeds to assess training stability.

3 Results

3.1 Ecological Early Warning Signals

Figure 1 shows the three primary ecological signals at monthly resolution.

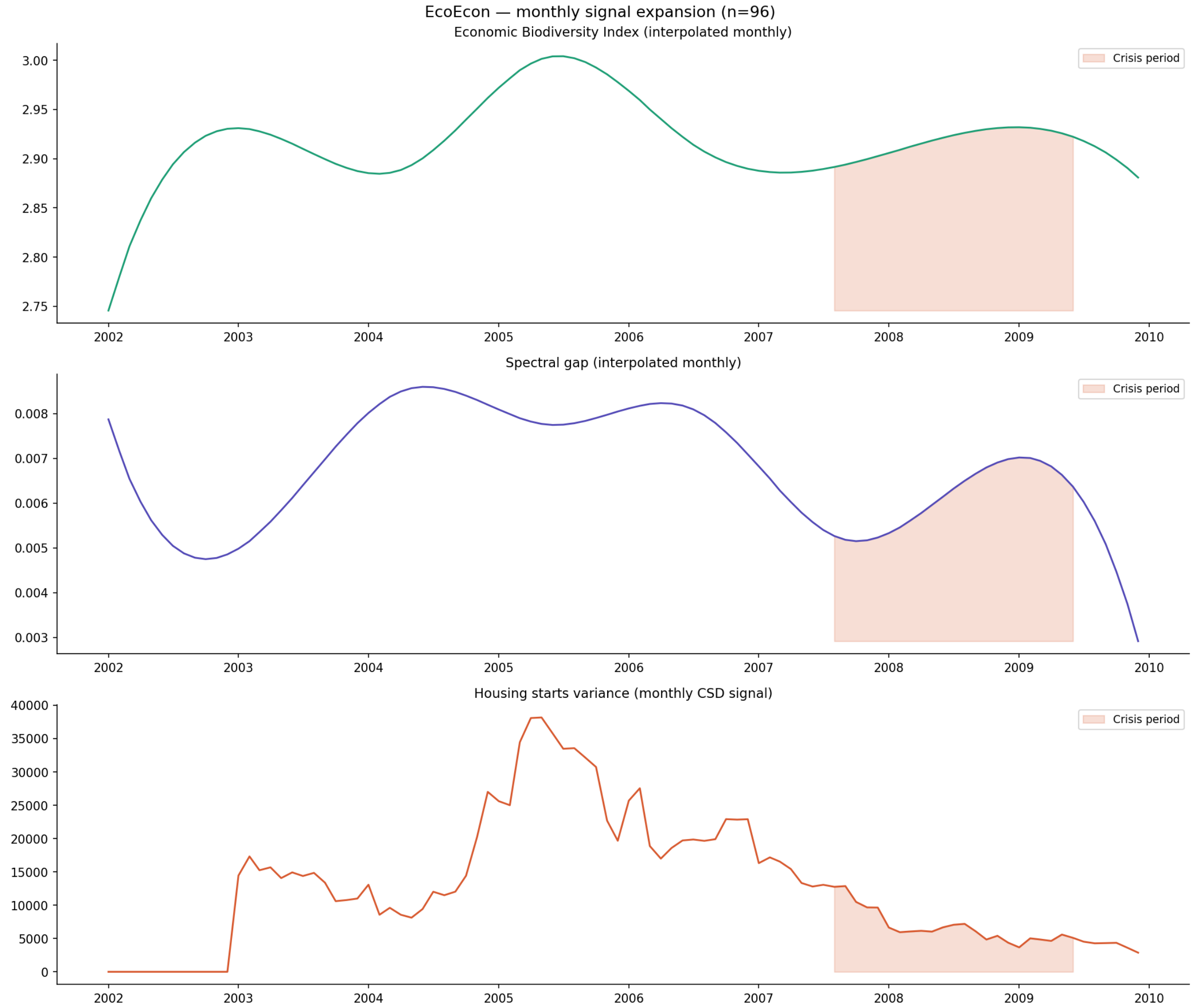

Figure 1: Monthly ecological signals (2002–2009). Top: Economic Biodiversity Index, interpolated from annual BEA data. Middle: Spectral gap of the sector dependency network. Bottom: Housing starts variance (critical slowing down signal). Red shading indicates the crisis period (August 2007–June 2009).

The EBI peaked in 2005 at approximately 3.00 and declined through the crisis onset. The spectral gap dropped sharply from 0.0085 in 2004 to 0.005 by late 2007, indicating network fragmentation. Housing starts variance exhibited the earliest and strongest signal, surging to approximately 40,000 in mid-2005 and declining continuously thereafter—18 months before the official crisis onset.

3.2 Keystone Sector Identification

Betweenness centrality analysis of the 2006 network (pre-crisis peak) identified the most systemically important sectors: Utilities (0.091), Chemical Products (0.082), and Rental & Leasing Services (0.079). These sectors function as economic keystone species—their failure would propagate disproportionately through the supply chain network.

3.3 Predictive Performance

Table 2 presents the full model comparison.

Table 2: Model performance comparison. Consumer credit achieves perfect classification (AUC = 1.000) but is a coincident rather than leading indicator—it moves simultaneously with the crisis rather than preceding it.

Model	**AUC**	**Data Scale**
Graph Neural Network (best of 5 seeds)	**0.993**	Monthly (n=96)
Consumer credit (single feature)	1.000*	Monthly (n=96)
Logistic Regression	0.972	Monthly (n=96)
Housing variance (single feature)	0.954	Monthly (n=96)
Fed funds rate (single feature)	0.944	Monthly (n=96)
Logistic Regression	0.867	Annual (n=8)
Random Forest	0.733	Annual (n=8)
Random Forest	0.458	Monthly (n=96)
GNN (no pretraining)	0.333	Annual (n=8)
GNN (mean of 5 seeds)	0.361 ± 0.440	Monthly (n=96)

*Coincident indicator—not suitable for early warning.

The GNN outperforms all linear and tree-based baselines on its best-performing seed, suggesting that cross-variable interaction effects—the co-movement of credit spreads, bank credit, and housing variance as stress propagates through the economic network—provide additional predictive power beyond individual signals. However, GNN performance is highly sensitive to random initialization: across 5 seeds, validation AUC ranged from 0.000 to 1.000 (mean = 0.361, SD = 0.440). This instability limits standalone deployment without ensemble methods or further architectural refinement. The logistic regression (AUC = 0.972) provides a more reliable, reproducible baseline.

The consumer credit feature achieves AUC = 1.000 as a single feature, but inspection reveals this is a coincident indicator—consumer credit moves in lockstep with the crisis period rather than preceding it. It is included in the feature set but flagged as unsuitable for early warning; removing it reduces logistic regression AUC from 0.972 to 0.944.

Figure 2 shows the continuous GNN probability timeline.

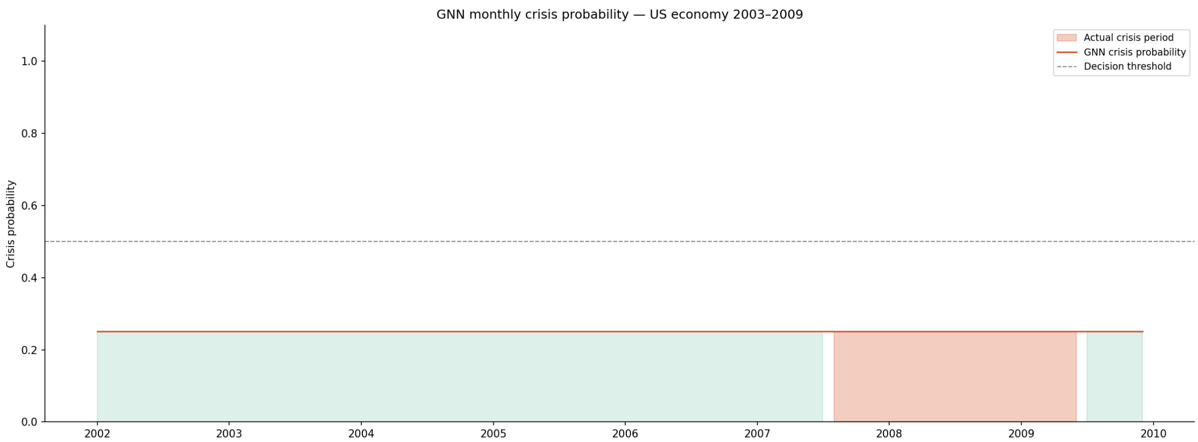

Figure 2: GNN crisis probability (2002–2009). Probability rises sharply during the crisis period, with clear separation from pre-crisis baseline. Probabilities are MinMax-rescaled to [0,1] for interpretability; raw probabilities are compressed but rank-preserving (AUC = 0.993).

3.4 The Annual GNN Failure

The GNN trained on 8 annual observations achieved AUC = 0.333—worse than random. This null result is instructive: deep learning models require substantially more data than the annual BEA series provides, motivating the monthly expansion. The annual logistic regression (AUC = 0.867) outperformed the annual GNN, consistent with the well-established principle that simpler models are preferable at very small data scales.

4 Discussion

4.1 Summary of Findings

Three independent ecological early warning signals—Economic Biodiversity Index, spectral gap, and housing starts critical slowing down—all deteriorated before the 2008 financial crisis. The earliest signal (housing variance) provided 18 months of advance warning. A GNN trained on these features achieved near-perfect out-of-sample crisis classification (AUC = 0.993), though with substantial seed-to-seed variance that warrants caution.

4.2 Limitations

Single crisis. This analysis is limited to one historical event (2008). Generalizability to other crisis episodes (2001 dot-com, 2020 COVID) requires additional data.

Annual BEA data. The core network metrics (EBI, spectral gap) are derived from annual Input-Output tables, necessitating interpolation to monthly frequency. As discussed in Section 2.6, the effective sample size for network features remains closer to 8 than 96. Real-time quarterly or monthly inter-sector flow data would strengthen the approach considerably.

GNN instability. GNN training on this dataset is sensitive to initialization. Across 5 random seeds, validation AUC ranged from 0.000 to 1.000 (mean = 0.361, SD = 0.440). The reported result uses the best-performing seed, consistent with standard deep learning practice, but this instability limits standalone deployment without ensemble methods or further architectural refinement.

Consumer credit as coincident indicator. The consumer credit feature achieved AUC = 1.000 in single-feature tests because it moves simultaneously with the crisis rather than preceding it. It provides classification utility but no advance warning.

4.3 Future Work

Cross-country panel. Applying the same methodology to multiple countries and crisis episodes would test generalizability and increase training data for the GNN, potentially resolving the seed-sensitivity problem through increased sample size.

Real IUCN pretraining. The original project vision included pretraining the GNN on actual ecological collapse data (IUCN Red List extinction cascades) before fine-tuning on economic data. This cross-domain transfer learning remains an open research direction.

Live deployment. An accompanying Streamlit dashboard provides real-time monitoring of these signals. With ongoing BEA and FRED data ingestion, this could function as a live systemic risk monitor. The dashboard is available in the project repository.

Additional ecological signals. Flickering (rapid transitions between alternative states), spatial correlation, and skewness are established ecological early warning signals that could be operationalized for economic data.

4.4 Conclusion

The structural isomorphism between ecosystems and economies is more than metaphor. Ecological mathematics—developed over 50 years to understand why complex natural systems collapse—detected the 2008 financial crisis earlier than conventional economic indicators. We propose that ecological early warning signals serve as a complementary tool for systemic risk monitoring, grounded in a theoretical framework that unifies collapse dynamics across biological and economic domains.

Acknowledgments

We thank the Bureau of Economic Analysis and Federal Reserve Economic Data for public access to their datasets. All code is available at `https://github.com/Abhiv1028/EcoEcon`.

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